

Root Duality in Neo-Constructionist Theory: Unifying Indexical Identity and Computational Anchoring

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1. Introduction

Research on roots within Neo-Constructionist approaches can be broadly classified into three perspectives:

(1) Scholars such as Harley (2014) argue that roots function as indices. In syntactic structures, Harley represents ROOTs as arbitrary numerical values to distinguish one root from another. He contends that a root may exhibit different phonological forms across various syntactic environments. Given the high variability of semantic content associated with roots—similar to semantic substitution effects—they cannot be identified solely by their phonological or semantic features. Thus, Harley proposes that roots be treated as purely structural units within the syntax, devoid of both phonological and semantic features, and used exclusively for syntactic computation.

(2) In contrast, Marantz (1997), Harley (2009a), and Embick (2004a), among others, attribute ontological properties to roots. For instance, Marantz categorizes roots into types such as object roots (e.g., entities), state roots (e.g., properties), and event roots (e.g., actions). Rappaport Hovav and Levin (2010) propose a classification of roots into “manner” and “result” types, where the semantic type of a root determines the syntactic behavior of the corresponding verb.

(3) A third perspective, advanced by Acedo-Matellán & Mateu (2015) and Borer (2013), holds that the semantic properties of roots are not inherent but rather derived from their syntactic configurations. Roots are considered abstract, category-neutral units whose interpretation is determined solely by their syntactic position in combination with functional heads (e.g., v, n).

To summarize, current theoretical approaches differ in how they conceptualize

roots: either as devoid of meaning and phonological content, or as having a kind of basic phonological and semantic identity. As noted above, Harley's position emphasizes the "indexical" nature of roots—lacking both phonological and semantic content. In contrast, Marantz (2024)¹ critiques the use of "indices" such as #53 or #67 in syntactic operations, arguing that these are not the elements manipulated in syntax. Instead, he maintains that the root must have a basic identity. For example, the morphological variation from "go" to "went" must be grounded in a root form, with indexical values merely serving as interface representations, not syntactic operands.

Regarding the claim that roots lack inherent meaning, Borer (2013), within the Exoskeletal Model (XS), asserts that roots carry no lexical semantics; their meanings are retrieved from encyclopedic knowledge at some point during the syntactic derivation.

From this review, it becomes clear that linguistic theories diverge not so much in their fundamental understanding of roots, but in their analytic emphasis. Therefore, the author proposes that the function of roots in language generation is multi-phased. Specifically, roots play a dual role during the mapping from the syntactic component to the interfaces—namely, the Sensory-Motor (S-M) and the Conceptual-Intentional (C-I) interfaces. In this view, roots function both as indexical units and as computational anchors.

This paper adopts the theoretical framework of Distributed Morphology (DM), particularly in its treatment of compounding, to argue for the dual nature of roots. The structure of the paper is as follows: Section 2 develops the hypothesis of root duality. Sections 3 and 4 examine the role of roots respectively as indices and as computational anchors in the formation of compound words. Section 5 provides a brief discussion on the implications of root duality for the theory of word formation. Section 6 concludes the paper. (Sections 3–6 will be updated later.)

2. The Proposal of Root Duality

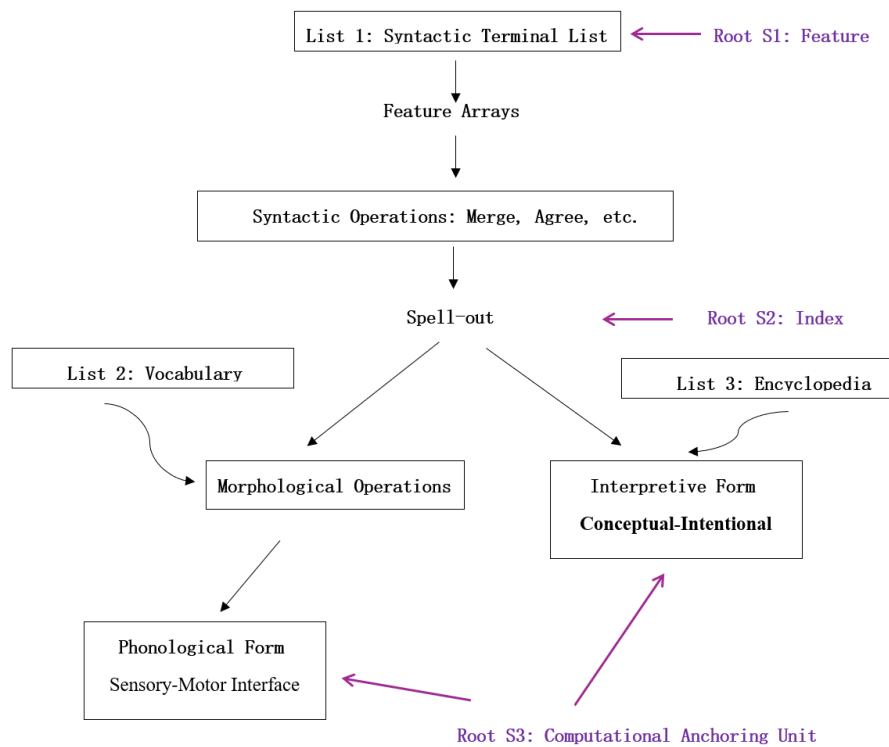
¹ Academic Correspondence between the Author and Alec Marantz.

This paper proposes that within the framework of Distributed Morphology (DM), roots (ROOTs) function as indices for interface identification. At the same time, their role is phase-dependent. During the syntactic operation phase—driven by List 1, which supplies terminal nodes for syntactic computation—the root is a non-phonological and non-semantic feature bundle. At this stage, the root merely serves as an indexical unit for interface identification, bearing no inherent phonological or semantic content.

However, during the morphological operation phase, also driven by List 1, which at this point provides Vocabulary Items for morphological insertion, the root undergoes ordered linearization. Here, the root acts as the base for morphological alternation. It is associated with a basic phonological form—determined post-syntactically—which serves as the anchor for phonological realization and morphological computation at the Phonological Form (PF) interface².

Similarly, at the encyclopedic phase, associated with List 3 (the Encyclopedia), the root functions as the semantic base for interpretation, with encyclopedic knowledge supplying context-sensitive meaning at the Conceptual-Intentional (C-I) interface.

² According to Marantz, morphological alternations such as the shift from “go” to “wen(t)” are initiated at the level of indexical representation, which presupposes the existence of an underlying phonological form for the root.



In summary, I argue that ROOTs are multi-phasal elements that exhibit a dual nature: they are simultaneously indexical units used for syntactic and interface reference, and computational bases for phonological and semantic interpretation. The unity of these two aspects underpins their central role in the grammar.